02

The KDD Process

The context

In recent decades, we have been witnessing a profound socio-economic transformation.
From an agricultural society, where the key resource is land, we had moved to an industrial one, where the key resources are energy and capital.
Now we are witnessing the transition to a post-industrial society, where the key resources are information and knowledge.
In this post-industrial society, information is no longer considered an accessory tool but a strategic asset, whose proper management can ensure the very survival of an organization or its differentiation from other competing players.

1st Industrial Revolution
- Mechanization, water power, steam power

2nd Industrial Revolution
- Mass production, assembly line, electricity

3rd Industrial Revolution
- Computers and automation

4th Industrial Revolution
- Cyber-physical systems, AI, IoT

The context

To describe this transformation, there are terms that have now become part of common language and that highlight the central role that information and knowledge play in many contexts:

- Information Society,
- Big Data,
- Data Science,
- Knowledge worker

There is growing awareness that the driving factor of innovation is the ability to manage information and supporting knowledge.

Data-driven innovation

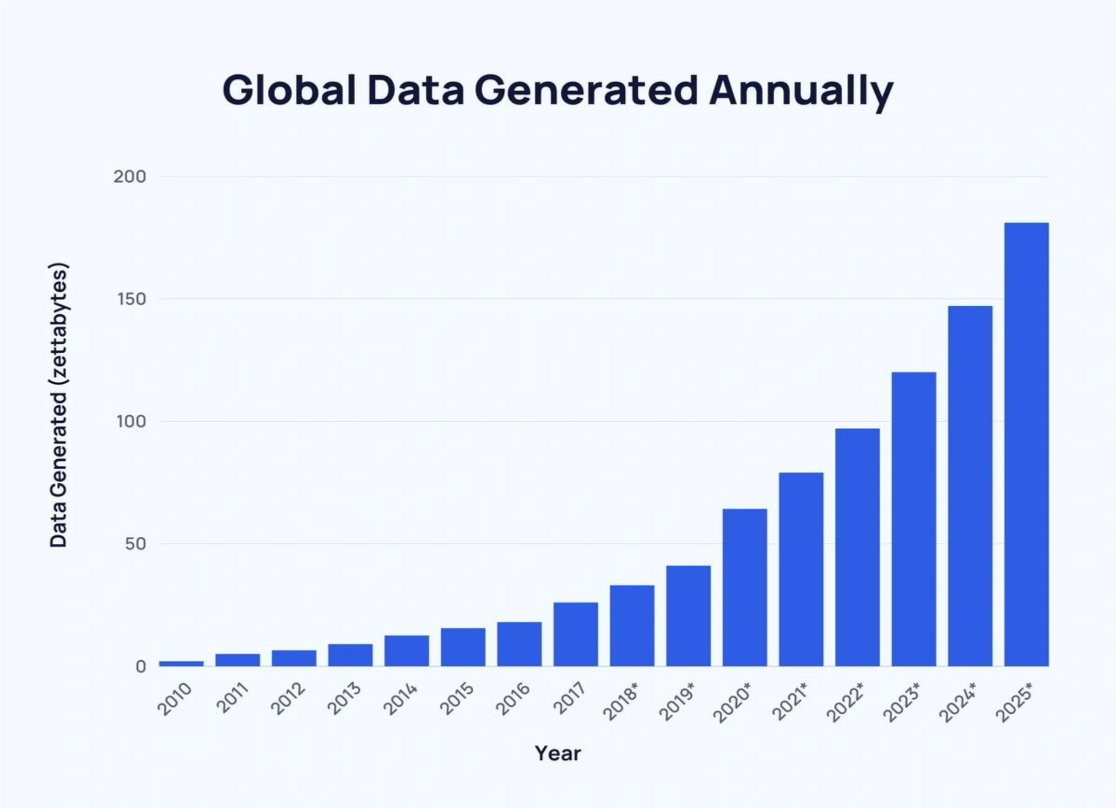
- The European Commission is pushing EU countries to embrace data-driven innovation.
- The term refers to the ability of organizations, businesses (including SMEs) or public administrations, to make use of information derived through computational data analysis tools to develop goods and services that improve the daily lives of both individuals and organizations themselves.

Data-driven innovation

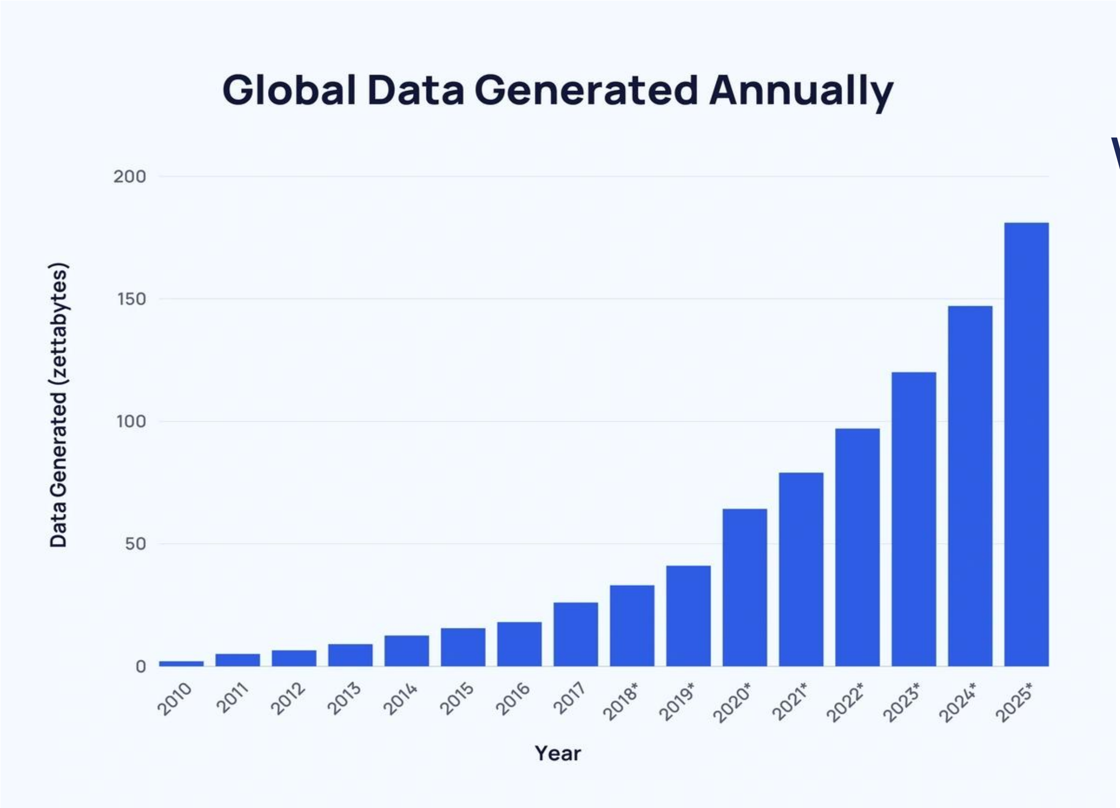
Data analytics techniques, when applied to large amounts of data (Big Data), increasingly allow organizations to:

- Personalize products/services,
- Optimize processes,
- Forecast market trends,
- Improve customer experience,
- Reduce decision-making risks.

Big Data

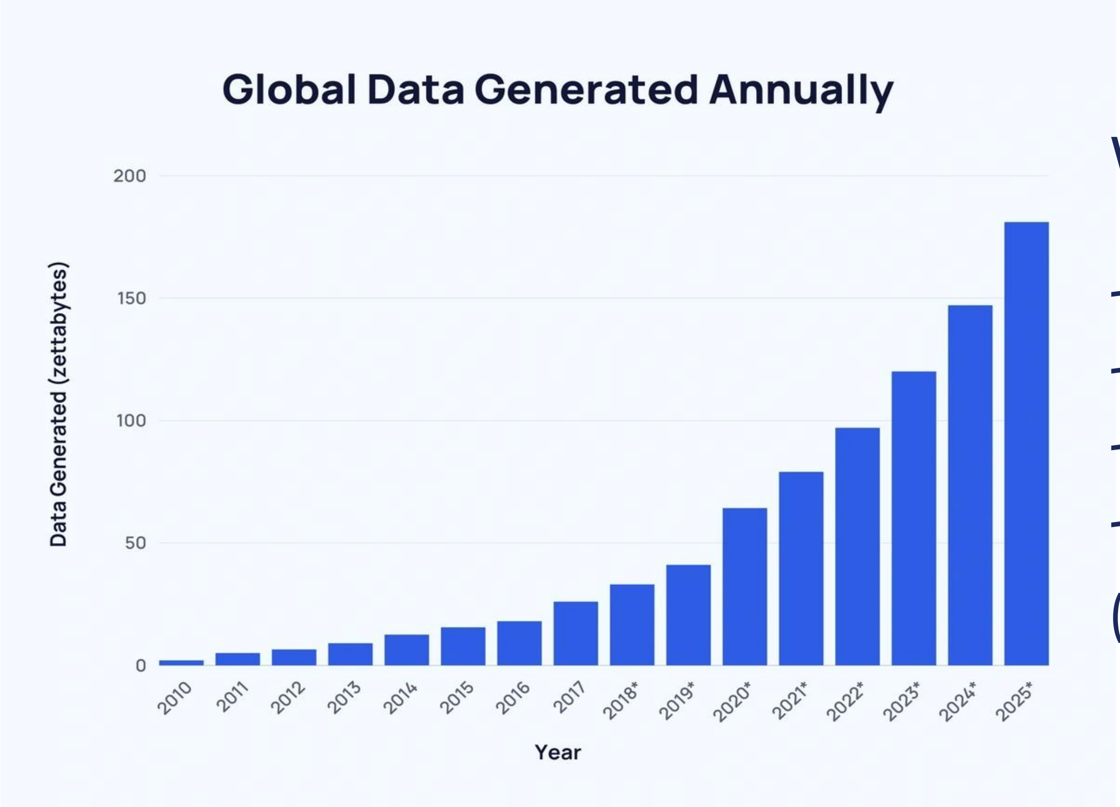

Global Data Generated Annually
Data Generated (zettabytes)
200
150
100
50
0
2010
2011
2012
2013
2014
2015
2016
2017
2018*
2019*
2020*
2021*
2022*
2023*
2024*
2025*
Year

Big Data



What is a zettabyte?

Big Data



What is a zettabyte?

10^{21} Byte
10^{15} Megabyte (MB)
10^{12} Gigabyte (GB)
10^{9} Terabyte (TB)
(one billion terabytes)

Big Data

WHAT'S A ZETTABYTE?

1 kilobyte	1,000,000,000,000,000,000,000
1 megabyte	1,000,000,000,000,000,000,000
1 gigabyte	1,000,000,000,000,000,000,000
1 terabyte	1,000,000,000,000,000,000,000
1 petabyte	1,000,000,000,000,000,000,000
1 exabyte	1,000,000,000,000,000,000,000
1 zettabyte	1,000,000,000,000,000,000,000

What is a zettabyte?

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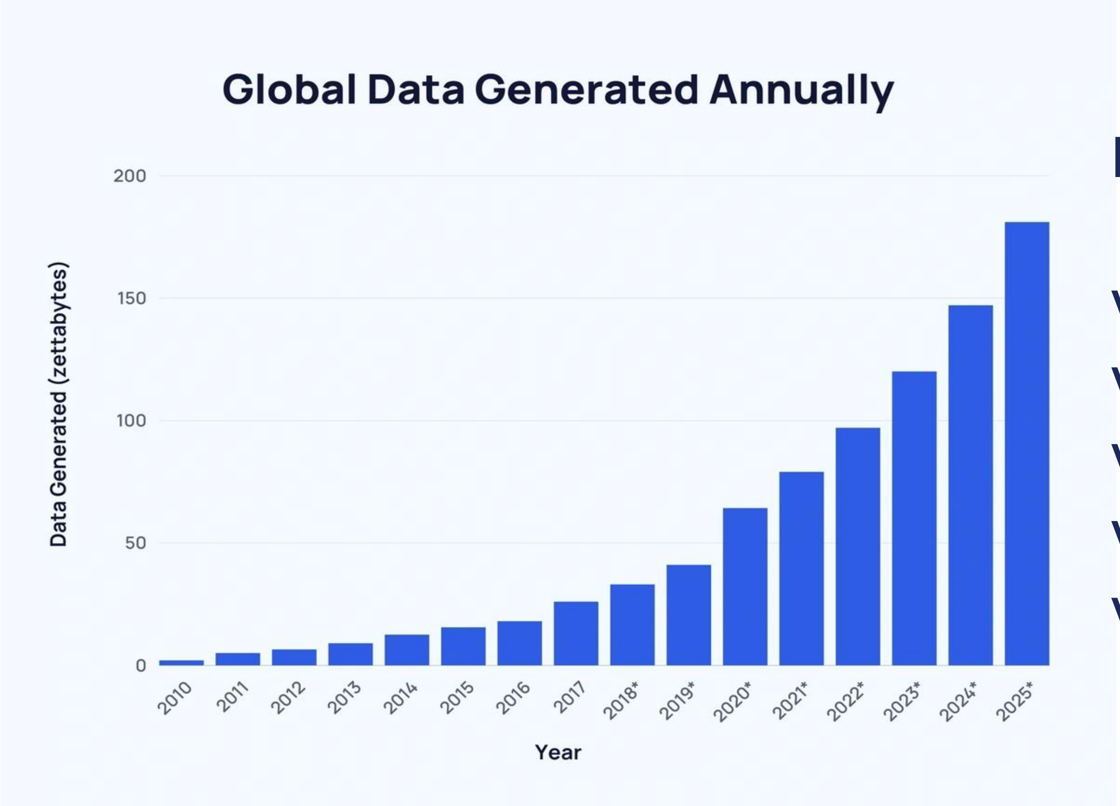
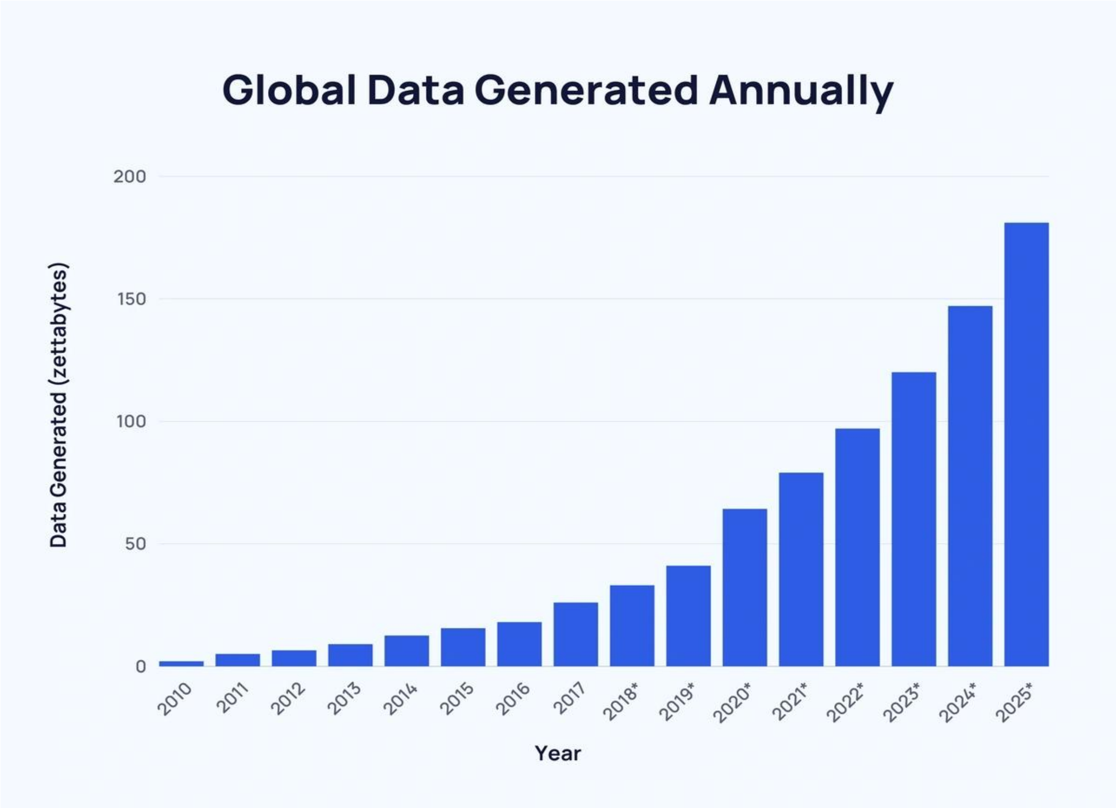
Big Data



Big Data five "V's"

Volume,
Velocity,
Variety,
Veracity,
Value

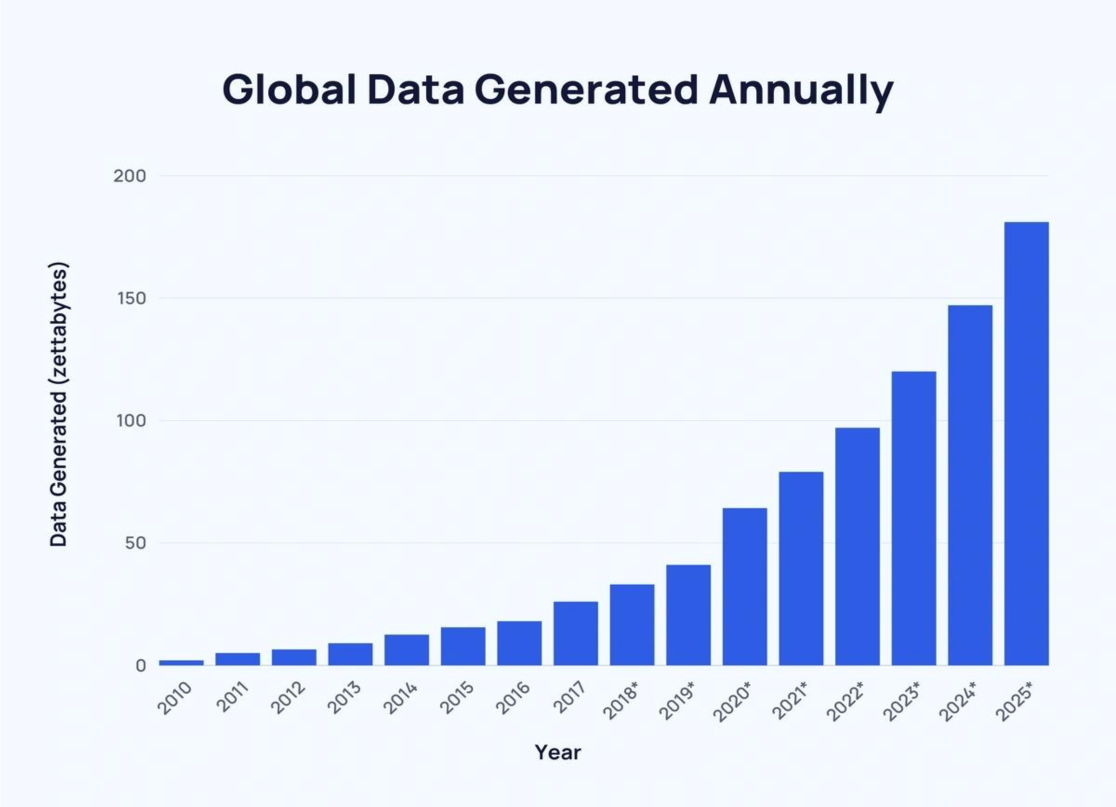
Big Data



Volume

It refers to the enormous amount of data generated and collected every day. This large quantity of data comes from sources such as sensors, social media, financial transactions, and so on.

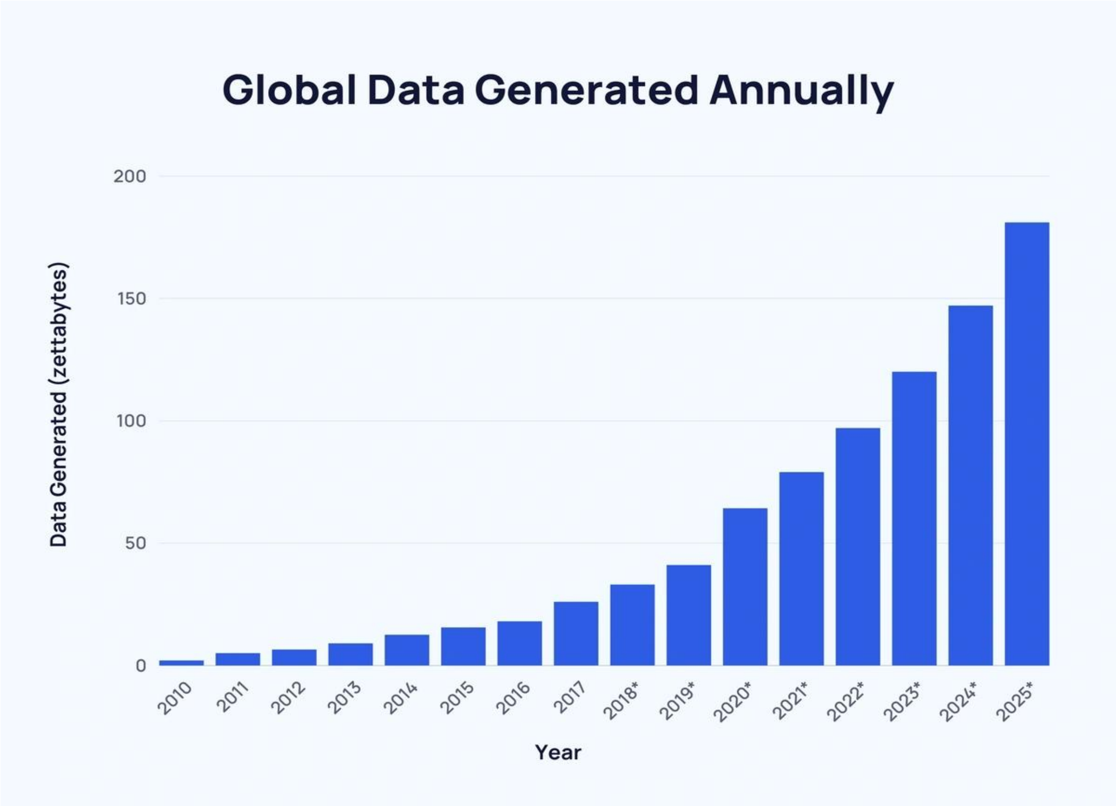
Big Data



Velocity

It refers to the speed at which data is generated and must be processed. With the advancement of technology, data is generated in real time and therefore it is essential to be able to analyze and leverage it quickly in order to make timely decisions.

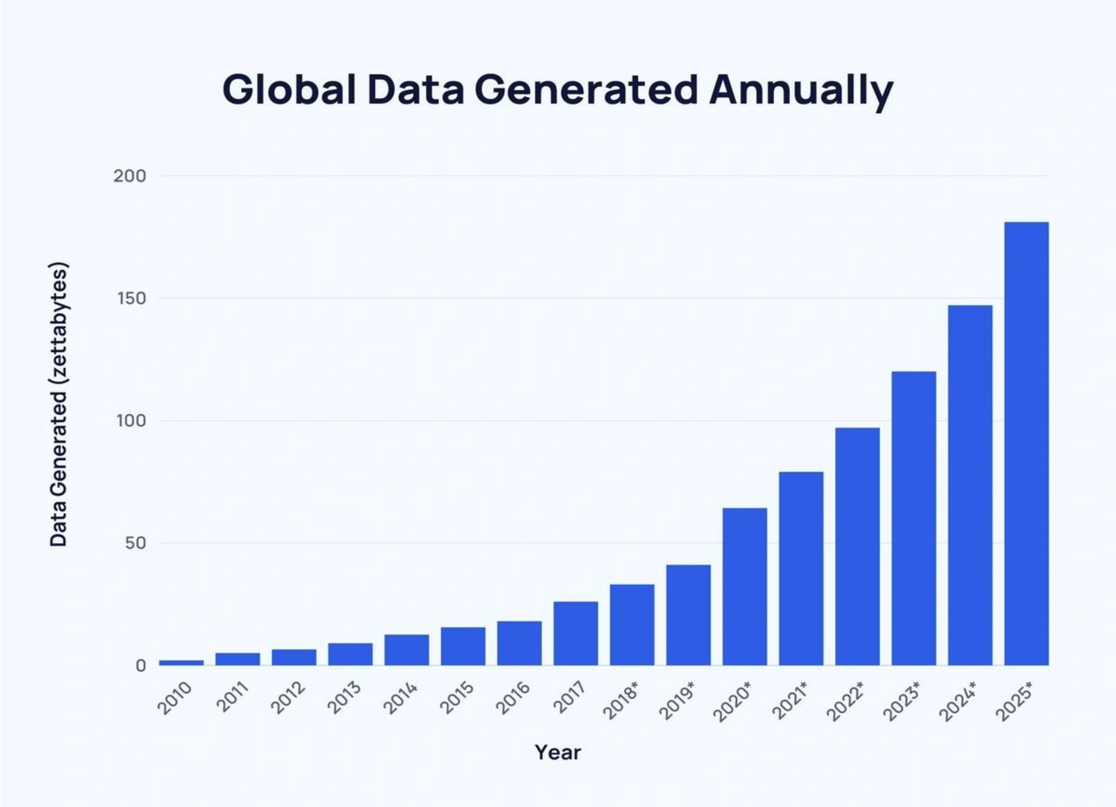
Big Data



Variety

Data comes from different sources and can be of many different types. This includes **structured** data (such as database tables), **semi-structured** data (such as JSON files) and **unstructured** data (such as images, videos, audio, text, etc.). Variety requires appropriate tools and methods to manage and analyze it..

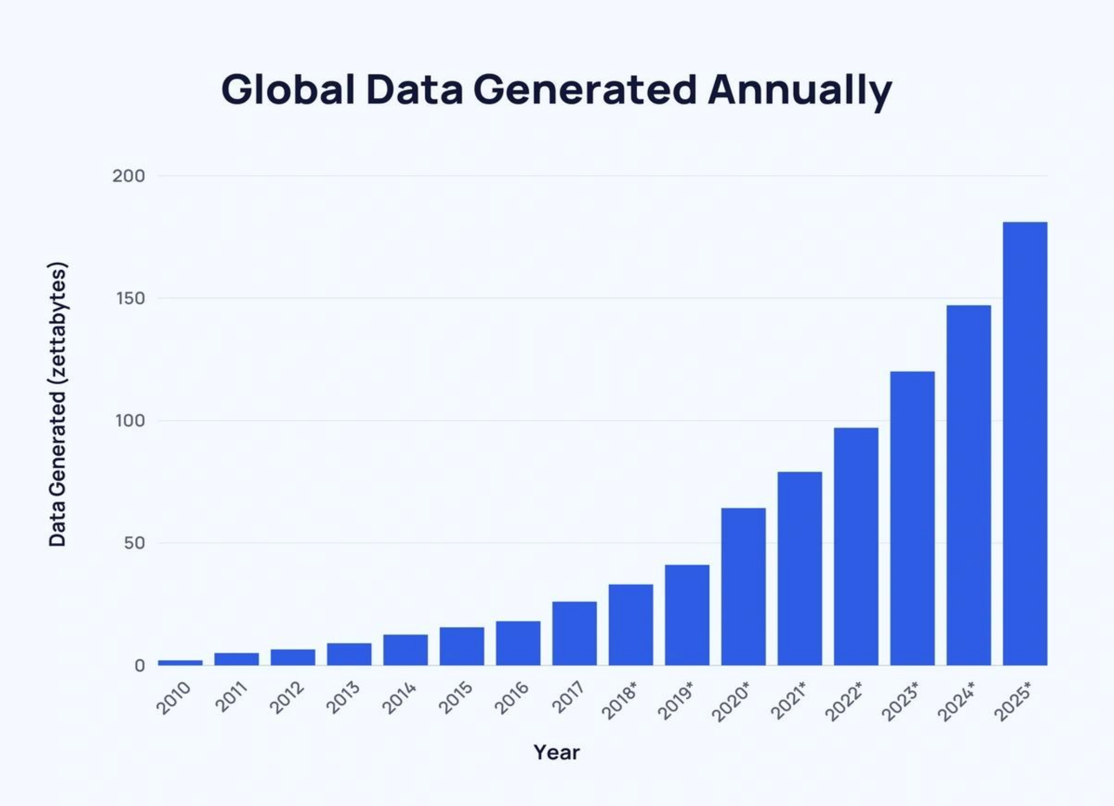
Big Data



Veracity

It represents the quality and reliability of data. Not all collected data is accurate, complete or consistent. Data may contain errors, duplications, or be incomplete, which can negatively impact analyses and decisions.

Big Data



Value

It is the ability to extract useful insights from data. Not all data is equally important or useful to an organization.

It is essential to be able to derive value from enormous amounts of data, extrapolating meaningful information to support strategic decisions and improve processes.

Data-driven innovation

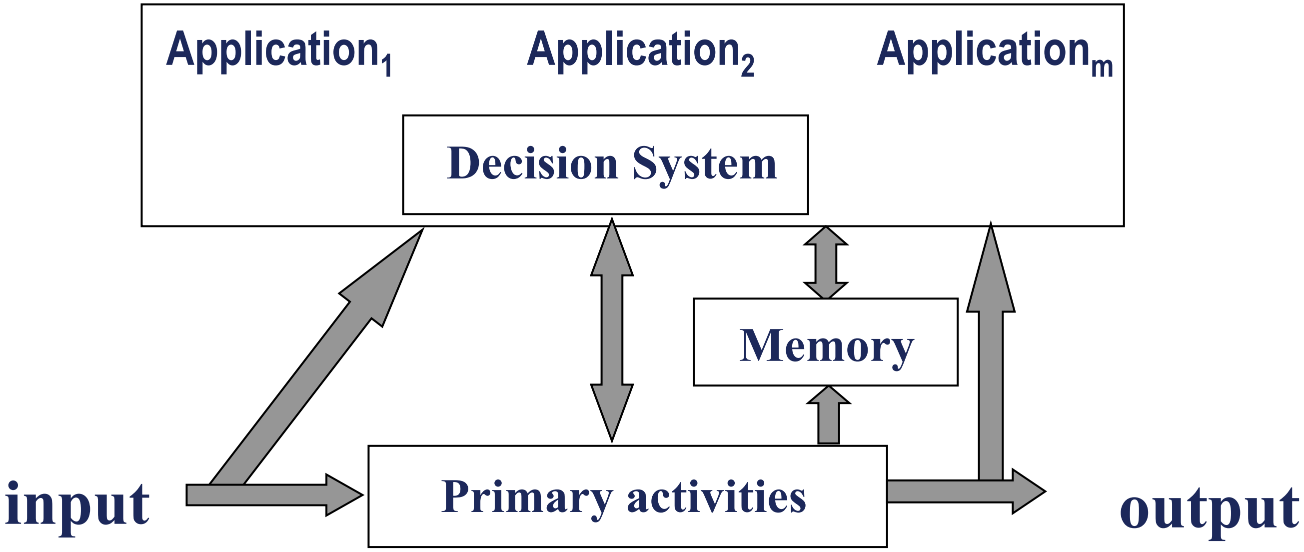
By "organization" or "organizational system" we mean the set of resources and rules that enable the functioning of any social structure in order to achieve its objectives.

Examples:

- A medical practice
- A library
- A manufacturing company
- A municipality

Data-driven innovation

The general principles of systems theory can be applied to the study of organizational systems, thereby identifying several components.



Data-driven innovation

In an organizational system there are:

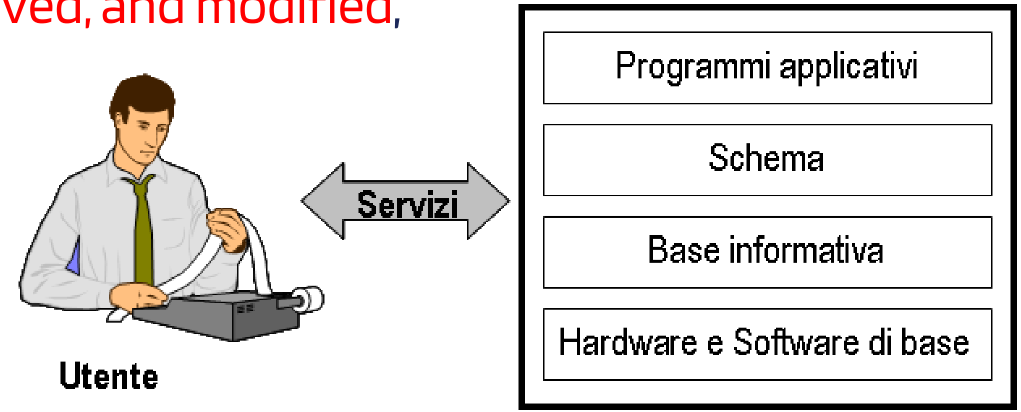
1. The operating system, which carries out primary activities.
2. The control and regulation system.
3. The information system, which ensures the acquisition, processing, circulation and storage of information.

Every organization (or organizational system) is therefore equipped with an information system, which organizes and manages the information necessary to pursue the goals of the organization itself.

Computer Systems

Thanks to the advent of ICT, it has been possible to automate a portion of the information system of organizations, producing computer systems.

More precisely, a computer system is a data management-oriented software system, in which the prevailing aspect is represented by the data that are stored, retrieved, and modified, and which constitute the information assets of an organization..



Computer Systems

While it is true that in an information society computer systems represent a crucial ICT technology for organizations, it is also true that the traditional orientation towards data management makes this technology inadequate with respect to the needs of an organization, which needs to access information and knowledge in order to support important decision-making processes.

Example:

The computer system that manages the student's academic career (enrollment, registration for subsequent years, regularity of fee payments, exams taken) is not helpful when it comes to deciding:

- Whether the distribution of credits in a degree program is reasonable
- What actions to take to improve the educational offering
- ...

Computer Systems

To understand the limitations of traditional computer systems, it is necessary to clarify:

- The concepts of data
- The concept of information
- The concept of knowledge

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What is data?

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What is data?

And what is information?

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- The concept of knowledge

What is data?

And what is information?

And what is knowledge?

Data

Tratto dalla Enciclopedia Zanichelli

dato s. m. Elemento o serie di elementi accertati e verificati che possono servire di base a ulteriori ricerche, indagini e sim. o che comunque consentono di giungere a determinate conclusioni | Enunciato o ipotesi su cui si basa una proposizione da dimostrare | (INF.) Informazione fornita in ingresso, memorizzata, elaborata o restituita in uscita in un processo di elaborazione, generalm. sotto forma di una sequenza di bit.

Data

From Cambridge dictionary

Data information, especially facts or numbers, collected to be examined and considered and used to help decision-making, or information in an electronic form that can be stored and used by a computer

Information and Knowledge

Tratto dalla Enciclopedia Zanichelli

informazióne s. f. (MAT.) Secondo la definizione di Shannon, tutto ciò che è in grado di ridurre l'incertezza tra diverse alternative possibili | (ELETTR., INF.) Ciascuno dei segnali trasmessi attraverso una via di comunicazione, il cui insieme forma il messaggio | In un sistema di elaborazione, ogni elemento associabile a una particolare configurazione del codice utilizzato (-> bit, -> byte).

conoscènza s. f. Cognizione, nozione già acquisita.

Information and Knowledge

From Cambridge Dictionary

information facts about a situation, person, event, etc.

knowledge understanding of or information about a subject that you get by experience or study, either known by one person or by people generally:

.

Data, Information, Knowledge

100

Data

It has no meaning

What is 100?

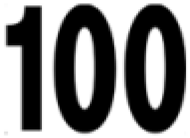
E.g. Temperature?

Measurement?

Price?

Uncertainty

Data, Information, Knowledge



Data

It has no meaning

What is 100?

E.g. Temperature?

Measurement?

Price?

Information

It has a meaning

What is 100?

It is the maximum speed at which I can drive on this road

Uncertainty → Certainty

Data, Information, Knowledge

A data is a raw element, which needs to be processed in order to acquire meaning and become information.

- Data: that which is immediately present to knowledge, before any processing;
- Information: the result of a process of elaboration and interpretation of data, which brings matter previously unknown to the recipient.

Data, Information, Knowledge

- Example: for a stock market operator, the articles read in a newspaper are information.

Data are known facts that can be recorded on some medium in a symbolic form.

- Example: the words written in newspaper articles.

Data alone have no meaning, but once properly interpreted and correlated they provide information, which allows us to enrich our knowledge of the world. The concept of information refers to its perceiver, its user, in other words to the context that gives it meaning.

Data, Information, Knowledge

Knowledge: Familiarity, awareness, or understanding gained through study, experience, and learning. The ability to understand and use information in a meaningful way.

Example:

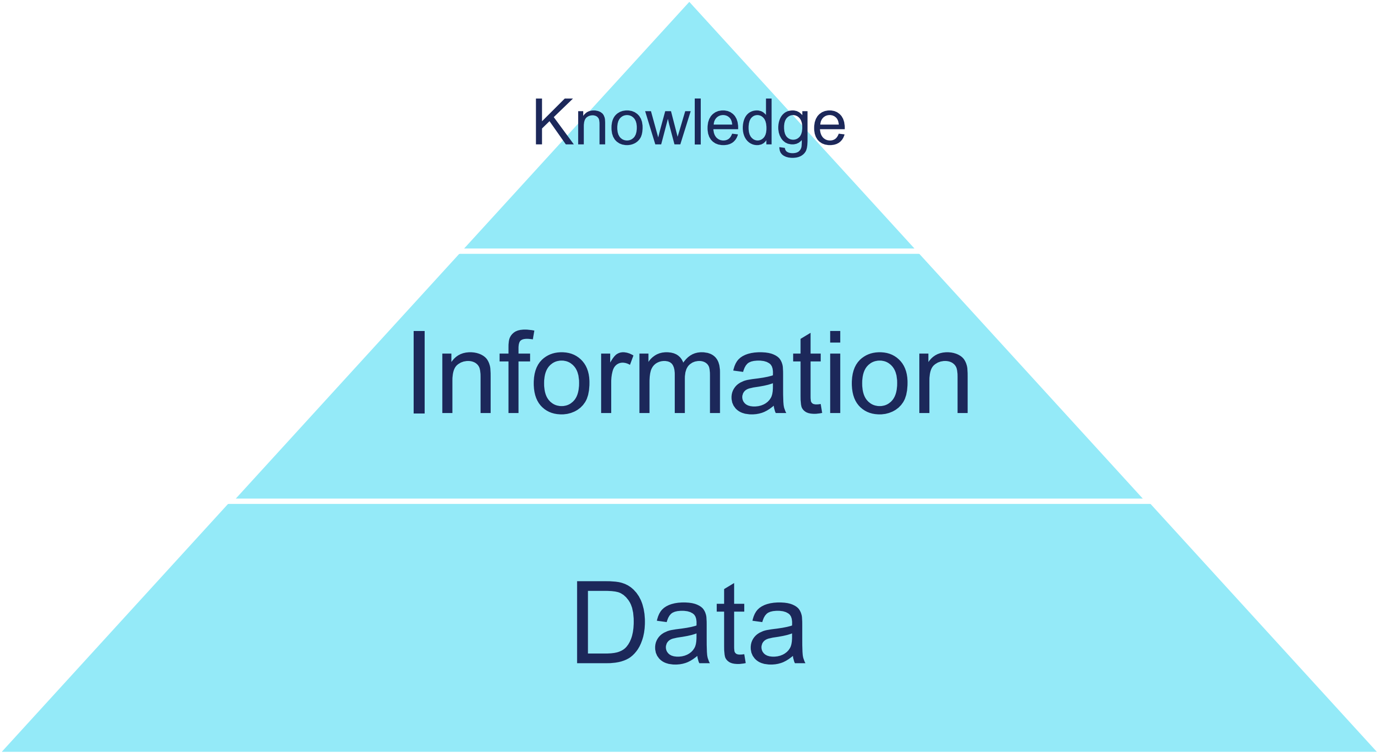
- Data consists of facts. For example, if you know the following data: **30°C**
- Information is data that takes on meaning in a given context: **The temperature in Rome is 30°C**
- Knowledge allows us to use information: **With 30°C in Rome in summer, it is advisable to go out wearing light clothing and protect yourself from the sun**

Data vs Knowledge

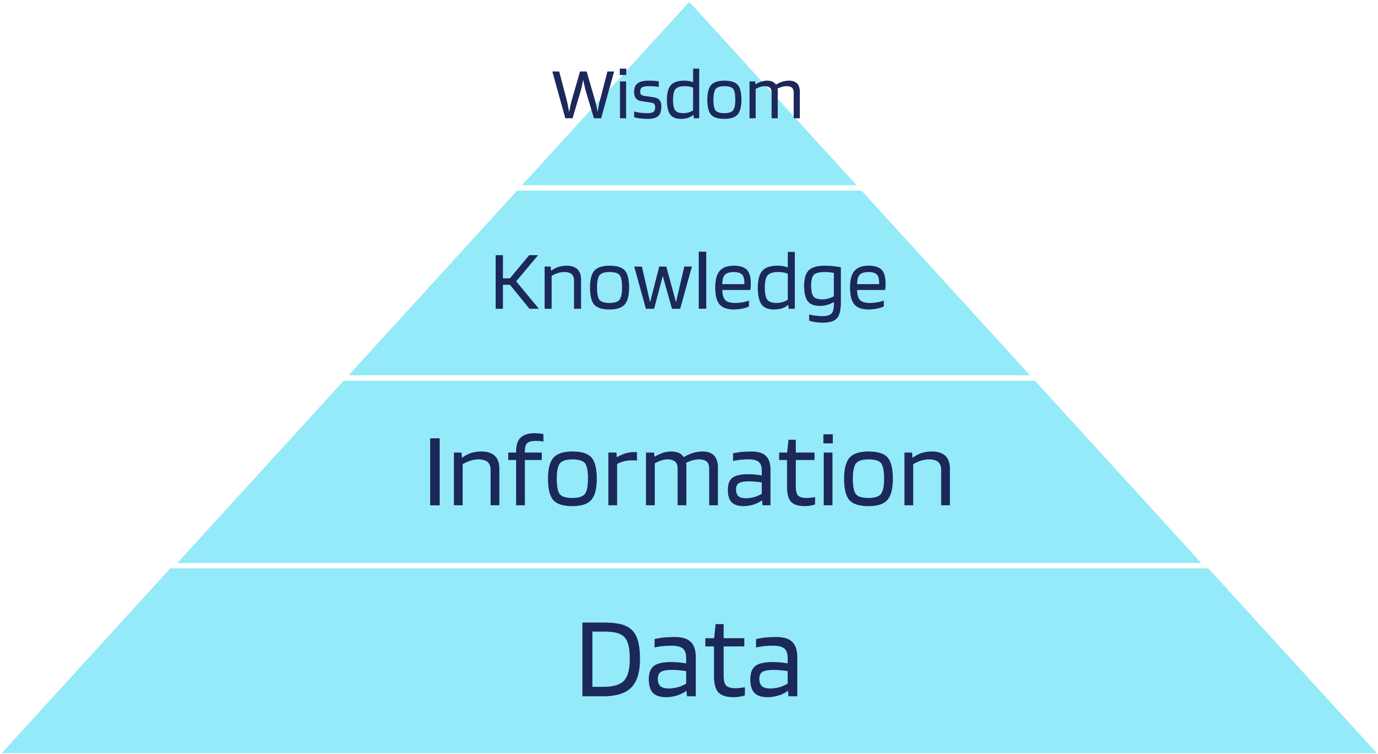
If we can rely on an automatic process or a simple clerk to collect material, then we are talking about data. The accuracy of data with respect to the real world can be objectively verified by comparison with repeated observations of the real world.

If we seek an expert to provide the material we need, then we are dealing with knowledge. Knowledge includes abstractions and generalizations of voluminous material. These are typically less precise and cannot be objectively verified.

Knowledge Pyramid

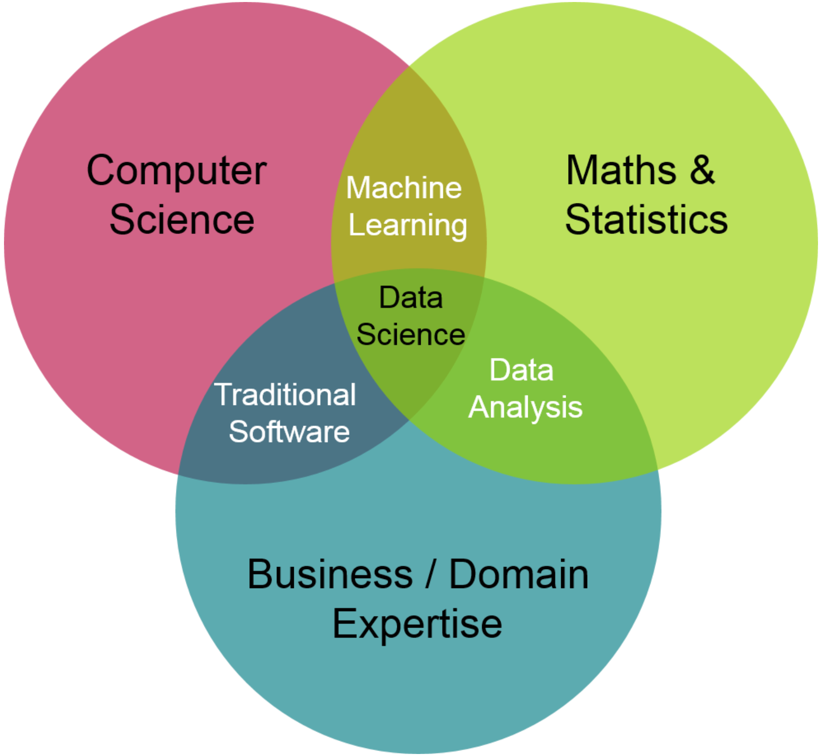


Extended Knowledge Pyramid - DIKW

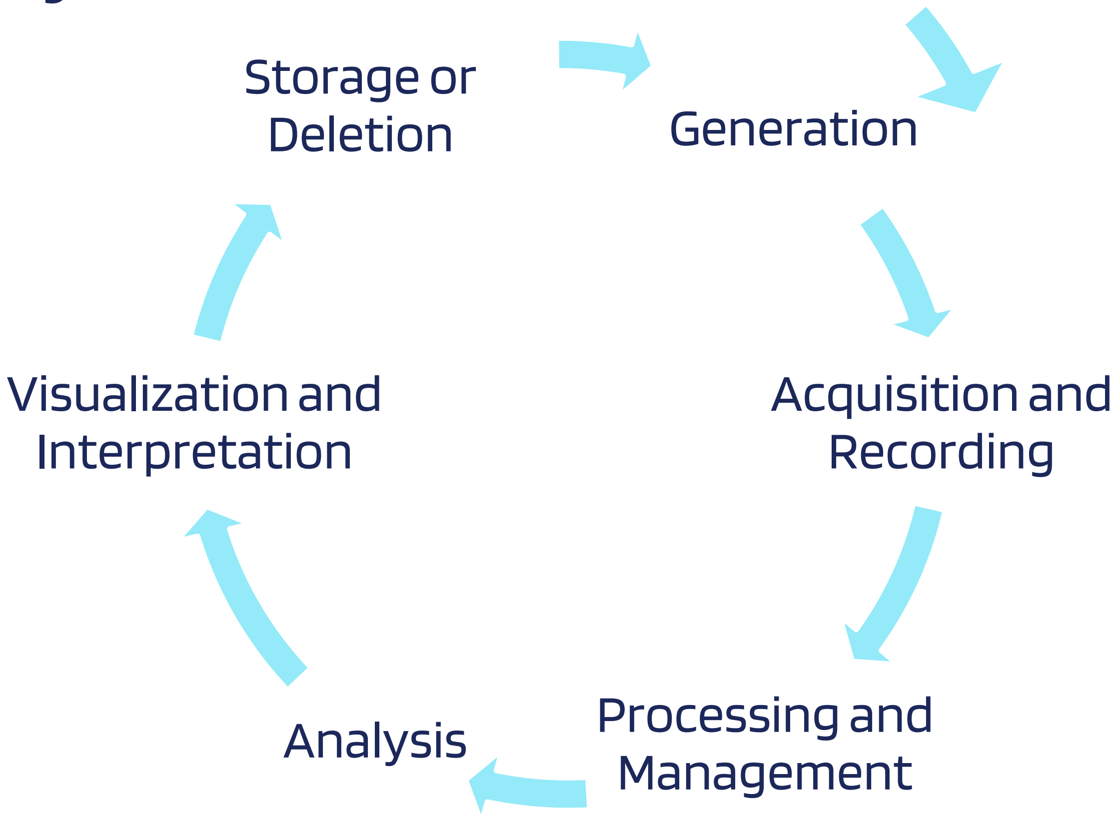


Data Science

Data science is the art and science of analyzing structured and unstructured data to identify patterns, make predictions, and take informed data-driven decisions.



Data lifecycle

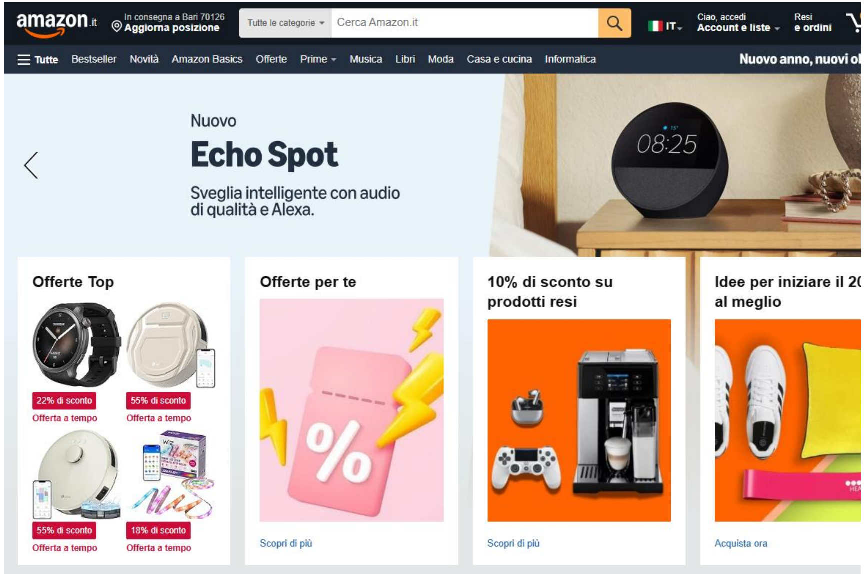


Data lifecycle

- Phase 1: Generation

Example: A user browses an e-commerce website
Generated data:

  - Product clicks
  - Time spent on the page
  - Products added to the cart
  - Purchases made
  - Reviews left

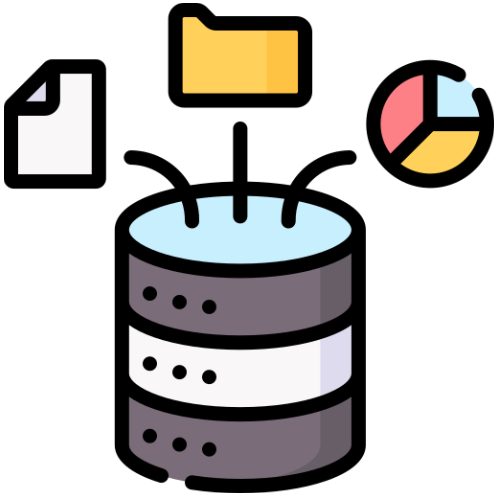


Data lifecycle

- Phase 2: Acquisition and Recording

Example: The system tracks and records user activities:

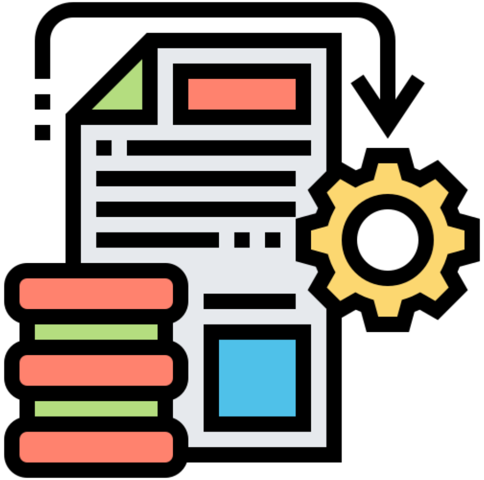
- Collection of browsing cookies
- Recording of purchases in the database
- Saving of user preferences
- Browsing session logsG
- DPR compliance: cookie consent and privacy



Data lifecycle

- Phase 3: Processing and Management

Example: Preparation of data for analysis
  - Cleaning of incomplete session data
  - Aggregation of purchases by category
  - Standardization of formats (dates, prices)
  - Integration of data from different sources (web, mobile app)
  - Identification and removal of duplicates



Data lifecycle

- Phase 4: Analysis

Example: Extraction of behavioral patterns

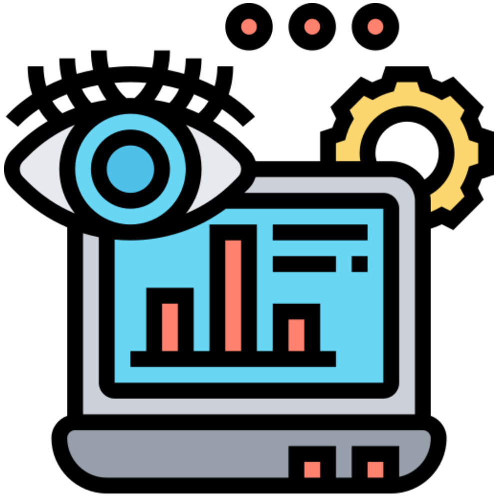
- Prediction of best-selling products
- Analysis of cart abandonment behavior
- Customer segmentation
- Identification of seasonal trends

Data lifecycle

- Phase 5: Visualization and Interpretation

Example: Dashboard for the marketing team

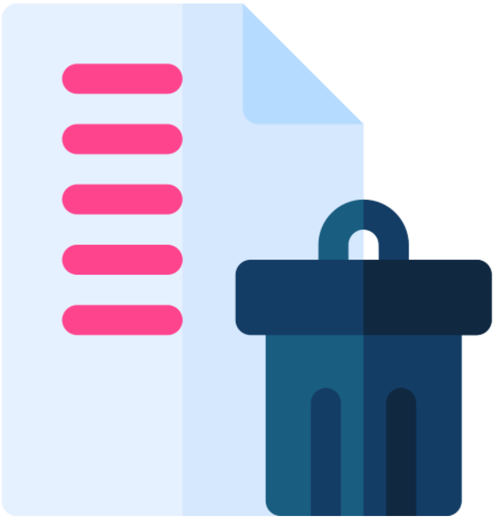
- Sales charts over time
- User behavior heatmap
- Automated weekly reports
- KPIs in real time
- Comparison with previous periods



Data lifecycle

- Phase 6: Storage or Deletion

Example: Historical data management and GDPR
  - Archiving of sales data for tax obligations
  - Deletion of inactive user data
  - Periodic encrypted backups
  - Historical data compression
  - Management of GDPR deletion requests



Knowledge Discovery in Data(bases)

We need a **mature methodology** that explains how large set of data can be analyzed.
This methodology has been investigated in a research area known as Knowledge Discovery in Data (KDD).

History: 1989, Gregory Piatetsky-Shapiro organized the first workshop on KDD @ IJCAI. The name of this research area originated there.
Goal of the workshop: investigating how **Machine Learning** (ML) techniques can be applied to **extract "knowledge"** from the huge mass of available data.
The workshop brought ML techniques to the attention of an immense set of potential users in all application domains

Knowledge Discovery in Data(bases)

The relationships among Machine Learning, Data Mining (which comes from Statistics) and Knowledge Discovery in Data Bases did not go without problems.

At the beginning, there was a confusion about the coverage of the terms. Now, the received view is that **KDD denotes the whole process of extracting knowledge, from data collection and pre-processing to results interpretation** (First KDD Conference, Montreal, Canada 1995).

Data Mining is the **step**, inside such a KDD process, in which information is actually extracted from the data by **applying algorithms** to them. These algorithms are often Machine Learning ones.

What is Knowledge Discovery

The most prominent, comprehensive attempt to define knowledge discovery as a process states:

Knowledge discovery is the nontrivial extraction of implicit, previously unknown, and potentially useful information from data.

Fayyad, U.M., Piatetsky-Shapiro, G., and Smyth, P. From Data Mining To Knowledge Discovery: An Overview. In Advances In Knowledge Discovery And Data Mining , eds. U.M. Fayyad, G. Piatetsky-Shapiro, P. Smyth, and R. Uthurusamy, AAAI Press/The MIT Press, Menlo Park, CA., 1996, pp. 1-34.

What is Knowledge Discovery

Knowledge discovery is the nontrivial extraction of *implicit*, previously *unknown*, and potentially *useful* information from data.

- *Nontrivial*: some search or inference is involved; that is, it is not a straightforward computation of predefined quantities (e.g., computing an average value).

What is Knowledge Discovery

Knowledge discovery is the nontrivial extraction of *implicit*, previously *unknown*, and potentially *useful* information from data.

- *Implicit*: information is implicit in the data and not explicitly formulated; explicit information is extracted by other techniques (e.g., database querying)

What is Knowledge Discovery

Knowledge discovery is the nontrivial extraction of *implicit*, previously *unknown*, and potentially *useful* information from data.

- *Unknown:* information should be novel; novelty depends on the assumed frame of reference. For instance:
  if Driver-Education-Course = Yes Then Age > 16
  is tautological for humans, but new for a machine.

What is Knowledge Discovery

Knowledge discovery is the nontrivial extraction of *implicit*, previously *unknown*, and potentially *useful* information from data.

- *Useful*: Information can help to achieve a goal of the system or the user. Patterns completely unrelated to current goals are of little use and do not constitute knowledge within the given situation.

A More Formal Definition

Given

- a set of ***facts*** (data) **F**,
- a representation language **L**,
- some measure of certainty C,

Define

a ***pattern*** as a statement ***S*** in ***L*** that describes relationships among a subset F_S of ***F*** with a certainty ***c***, such that ***S*** is simpler (in some sense) than the enumeration of all facts in F_S.

A pattern is considered to be ***knowledge*** if it is *interesting* and *certain* enough (or *valid*).

A More Formal Definition

Patterns and languages: in KDD we are interested in patterns that are expressed in a high-level language, such as:

If Age $<$ 25 and Driver-Education-Course = No

Then At-fault-accident = Yes

With Likelihood = 0.2 to 0.3

Such patterns can be understood and used directly by people, or they can be the input to another program (e.g., an expert system or a semantic query optimizer).

A More Formal Definition

Certainty: Without sufficient certainty, patterns become unjustified and, thus, fail to be knowledge. Certainty involves several factors:

- integrity of data
- size of sample on which the discovery was performed
- the degree of support from available domain knowledge.

Interestingness: Patterns are interesting when they are:

- Novel
- Useful
- Nontrivial to compute

Patterns vs. Models

What is the difference between patterns and models?

Patterns vs. Models

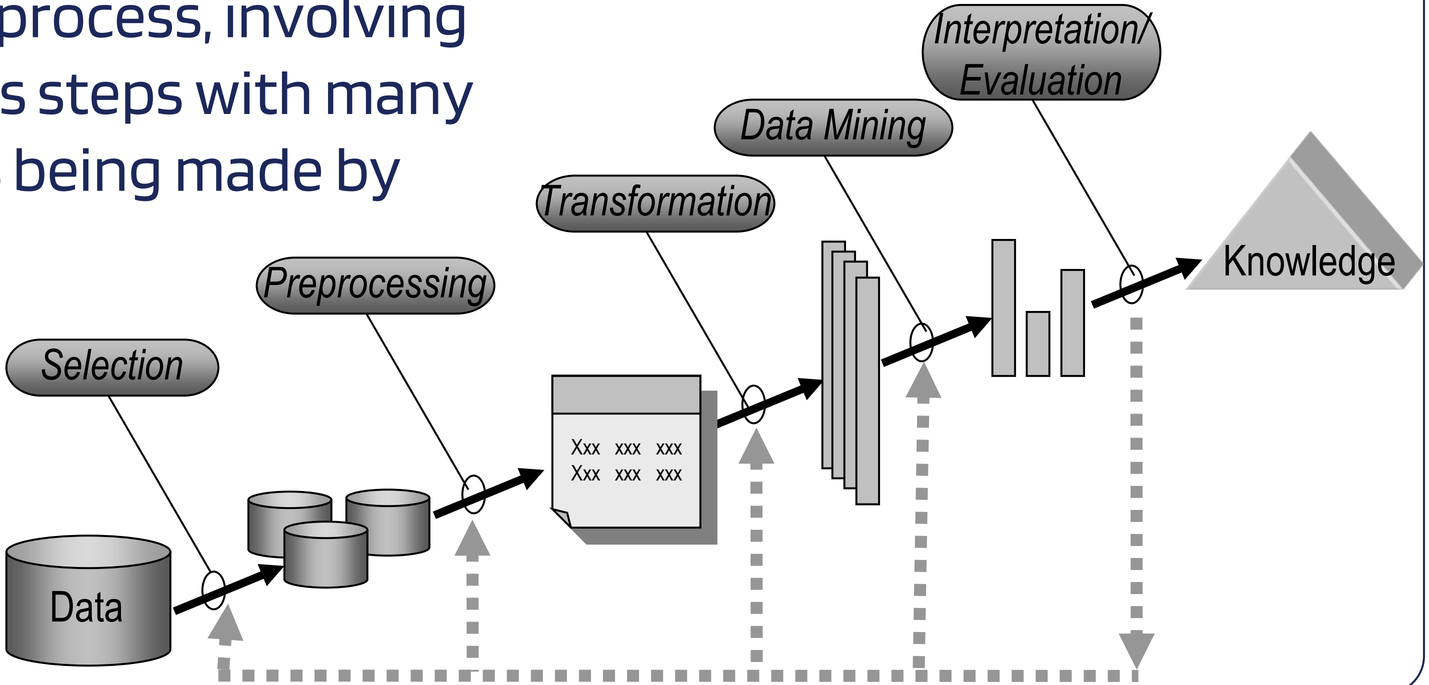
Patterns vs. Models: different interpretations

1st *interpretation*: a model is a global summary of the dataset, while a pattern is a local feature of the dataset (limited to a subset of observations and / or attributes).

2nd *interpretation*: Data mining involves fitting *models* to, or determining *patterns* from, observed data. A ***pattern*** can be thought of as instantiation of a ***model***, e.g., $f(x)=3x^2+x$ is a pattern whereas $f(x) = \alpha x^2 + \beta x$ is considered a model. The fitted models play the role of inferred knowledge.

The KDD Process: An Overview

- KDD is an **interactive** and **iterative** process, involving numerous steps with many decisions being made by the user.



The KDD steps

- Developing an ***understanding*** of the application domain, the relevant background knowledge, and the goals of the end-user.
- Creating a ***target data set***: selecting a data set, or focusing on a subset of variables or data samples, on which discovery is to be performed.
- ***Data cleaning and preprocessing***: basic operations such as the removal of noise or outliers if appropriate, collecting the necessary information to model or account for noise, deciding on strategies for handling missing data fields, accounting for time sequence information and known changes.

The KDD steps

- ***Data reduction and projection***: finding useful features to represent the data depending on the goal of the task. Using multidimensionality reduction or transformation methods to reduce the effective number of variables under consideration or to find invariant representations for the data.
- ***Choosing the data mining task***: deciding whether the goal of the KDD process is classification, regression, clustering, etc.

The KDD steps

- ***Choosing the data mining algorithm(s)***: selecting method(s) to be used for searching for patterns in the data. This includes deciding which models and parameters may be appropriate (e.g., models for categorical data are different than models on vectors over the reals) and matching a particular data mining method with the overall criteria of the KDD process (e.g. the end-user may be more interested in understanding the model than its predictive capabilities).

The KDD steps

- ***Data mining***: searching for patterns of interests in a particular representational form or a set of such representations: classification rules or trees, regression, clustering, and so forth. The user can significantly aid the data mining method by correctly performing the preceding steps.
- ***Interpreting mined patterns***, possible return to any of steps 1–7 for further iteration.
- ***Consolidating discovered knowledge***: incorporating this knowledge into the performance system, or simply documenting it and reporting it to interested parties. This also includes checking for and resolving potential conflicts with previously believed (or extracted) knowledge.

Standardizing the KDD process

Most work on KDD has focussed on step 7 — the ***data mining***.

However, the other steps are of considerable importance for the successful application of KDD in practice.

The need for a standardization of the knowledge discovery process has lead to the definition of the industrial standard **CRISP-DM** *(CRoss Industry Standard Process for Data Mining)*.

http://www.crisp-dm.org/

The CRISP-DM Process Model

Aim:

- To develop an industry, tool and application neutral process for conducting Knowledge Discovery
- Define tasks, outputs from these tasks, terminology and mining problem type characterization

Founding **Consortium Members**: DaimlerChrysler, SPSS and NCR

CRISP-DM Special Interest Group (SIG) > 300 members

- Management Consultants
- Data Warehousing and Data Mining Practitioners

CRISP 1.0 released in 1999

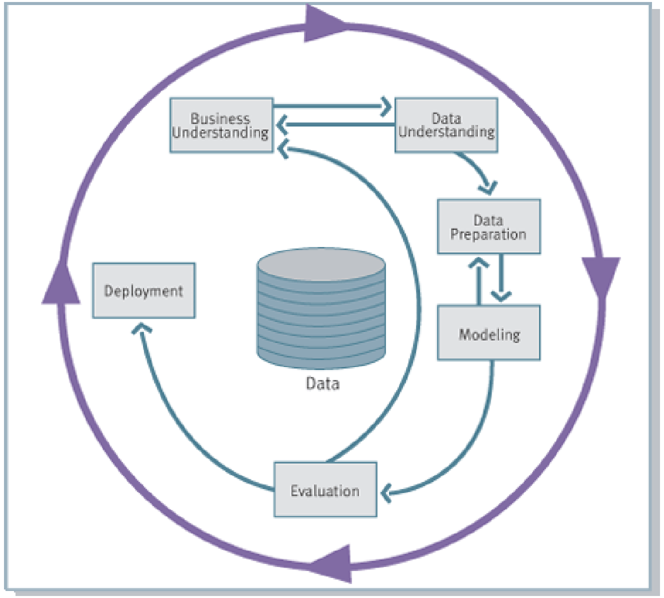
CRISP 2.0 SIG has been formed @ Chicago in Sept 2006.

The CRISP-DM Process Model

The CRISP-DM view on knowledge discovery consists of six phases.

The sequence of the phases is not strict. It is possible to move back and forth between different phases depending on the outcome of each phase. The arrows indicate the most important/frequent dependencies.

The outer circle symbolizes the cyclic nature of a KDD process which can continue after a solution has been deployed.

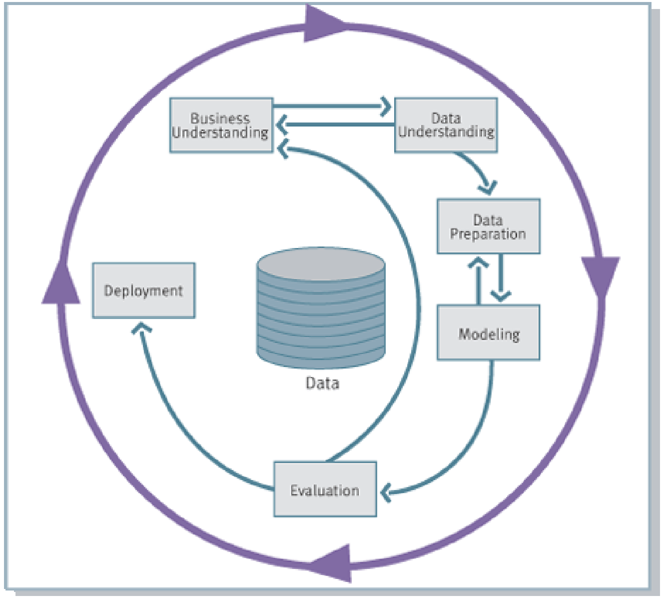


Life cycle of a KDD project

The CRISP-DM Process Model

Each phase contains a number of tasks which produce specific outputs.

Next slides characterize these tasks and outputs.



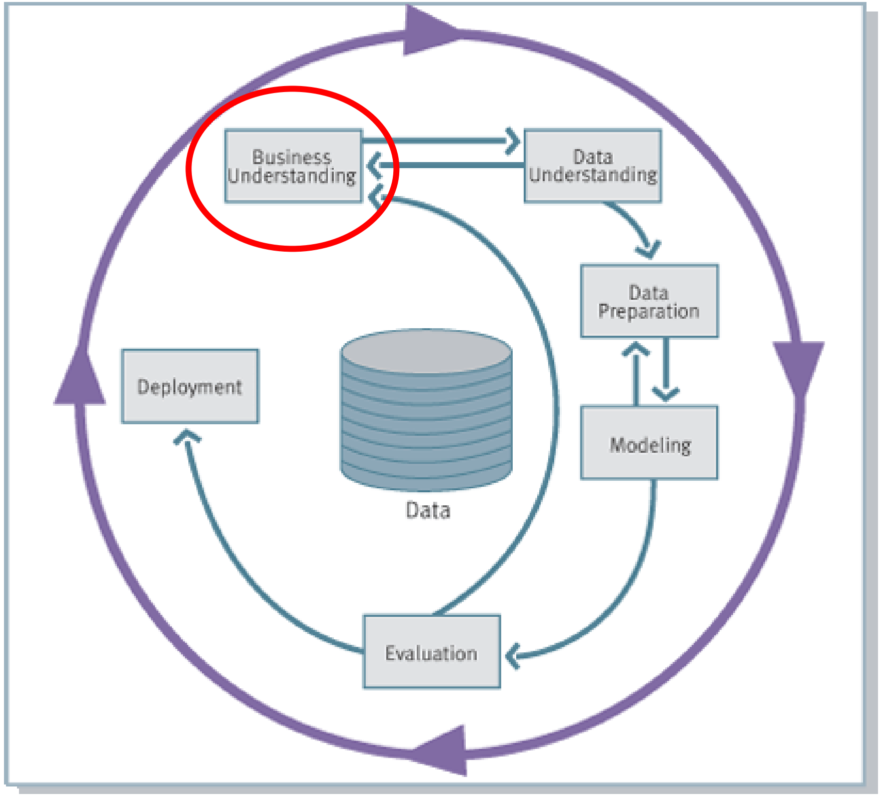
The CRISP-DM Process Model

NCR

Phases and Tasks

Business Understanding	Data Understanding	Data Preparation	Modeling	Evaluation	Deployment
Determine Business Objectives *Background* *Business Objectives* *Business Success Criteria*	**Collect Initial Data** *Initial Data Collection Report*	*Data Set* *Data Set Description*	**Select Modeling Technique** *Modeling Technique* *Modeling Assumptions*	**Evaluate Results** *Assessment of Data Mining Results w.r.t. Business Success Criteria* *Approved Models*	**Plan Deployment** *Deployment Plan*
Situation Assessment *Inventory of Resources* *Requirements, Assumptions, and Constraints* *Risks and Contingencies* *Terminology* *Costs and Benefits*	**Describe Data** *Data Description Report*	**Select Data** *Rationale for Inclusion / Exclusion*	**Generate Test Design** *Test Design*	**Review Process** *Review of Process*	**Plan Monitoring and Maintenance** *Monitoring and Maintenance Plan*
Determine Data Mining Goal *Data Mining Goals* *Data Mining Success Criteria*	**Explore Data** *Data Exploration Report*	**Clean Data** *Data Cleaning Report*	**Build Model** *Parameter Settings* *Models* *Model Description*	**Determine Next Steps** *List of Possible Actions* *Decision*	**Produce Final Report** *Final Report* *Final Presentation*
Produce Project Plan *Project Plan* *Initial Asessment of Tools and Techniques*	**Verify Data Quality** *Data Quality Report*	**Construct Data** *Derived Attributes* *Generated Records*	**Assess Model** *Model Assessment* *Revised Parameter Settings*		**Review Project** *Experience Documentation*
		Integrate Data *Merged Data*			
		Format Data *Reformatted Data*			

Business Understanding



Business Understanding

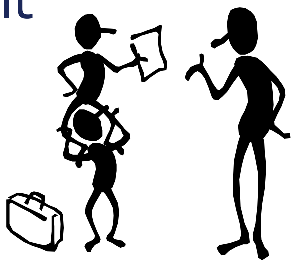
Determine Business Objectives

This step in the KDD process has a lot in common with the initial step of any significant project undertaking.

Minimum *requirements*:

- A perceived business problem or opportunity.
- Some level of executive sponsorship.

Developing an understanding and careful definition of the business needs is not a straightforward task in general. It requires the collaboration of the ***business analyst*** and the ***data analyst***



Business Understanding

Business Objectives Determination

This step in the process is also the time at which to start setting *expectations*.

Example.

Data mining projects to improve the response rates to mailing campaigns typically have objectives to increase the rate from the current 1% to 2%–2.5%.

Expectations of returns that are significantly above these levels will probably not be possible and will most likely doom the project to failure.

Business Understanding

Business Objectives Determination

OUTPUT:

- The background, which details information known about the business situation at the beginning of the process.
- The business objectives, which describe primary goals from business perspectives.
- The business success criteria, which define measures for sufficiently high-quality results of the project from business points of view.

Business Understanding

Assess situation

This task details background information on

- Resources
- Constraints
- Assumptions

considered in determining the data mining goals and the project plan

Business Understanding

Assess situation

OUTPUT:

- Inventory of
  - ✓ resources: personnel, data, computing & sw
  - ✓ constraints: schedule of completion, legal issues, comprehensibility
  - ✓ assumptions: data availability
- A glossary of terminology, which covers both business and data mining terminology.
- A cost-benefit analysis: expenditures of the project should be compared to potential gains.

Business Understanding

Determine Data Mining Goals

Business objectives (stated in business terms) have to be transformed into data mining goals (stated in technical terms).

OUTPUT:

- Data mining goals
- Data mining success criteria (e.g., a certain level of predictive accuracy).

Business Understanding

Determine Data Mining Goals

Typical data mining goals include:

- Classification
- Estimation
- Prediction

Directed Data Mining

Or

Predictive Data Mining

Or

Supervised Learning

- Affinity grouping
- Clustering
- Description & Profiling

Undirected Data Mining

Or

Descriptive Data Mining

Or

Unsupervised Learning

Business Understanding

Determine Data Mining Goals

The two "high-level" primary goals of data mining in practice tend to be:

- Prediction, which involves using some *independent* variables to predict unknown or future values of other *dependent* variables.

  IF Smoke=high AND Air-pollution=high THEN Disease=lung-cancer

- Description, which makes no distinction between dependent/independent variables and focuses on finding human-interpretable patterns describing the data.

  Current sales are up 20% over last year's sales

Data Mining Tasks

Classification

Classification is learning a function that maps (classifies) a data item into one of several predefined classes.

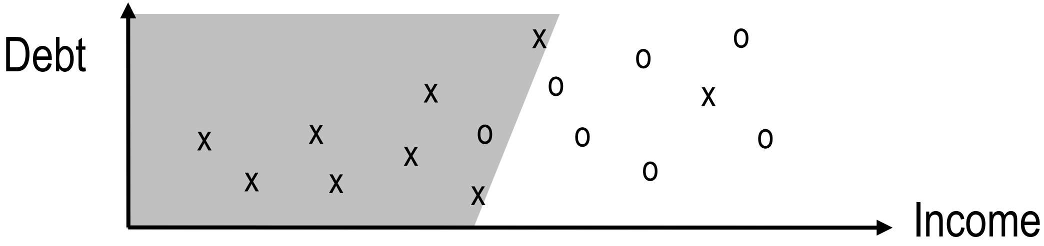
Applications

Automated classification of objects in large image databases

Other?

Data Mining Tasks

Classification



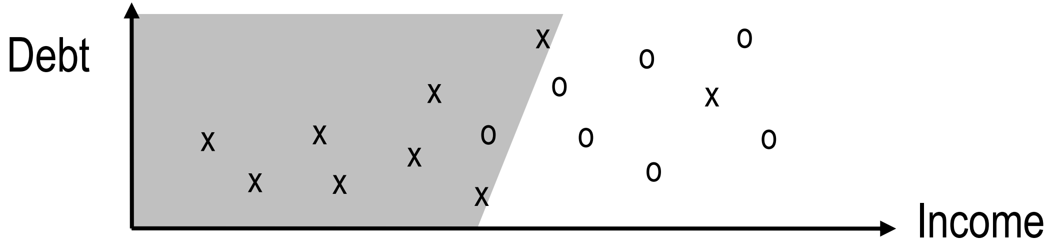
Example:

• Each point on the graph represents a person who has been given a loan by a particular bank at some time in the past.

• The horizontal axis represents the income of the person, the vertical axis represents the total personal debt of the person (mortgage, car payments, etc.).

Data Mining Tasks

Classification



Example:

The data has been classified into 2 classes:

- the x's represent persons who have defaulted on their loans,
- the o's represent persons whose loans are in good status with the bank.

The bank is interested to find a partition of the space into two regions.

Data Mining Tasks

Regression

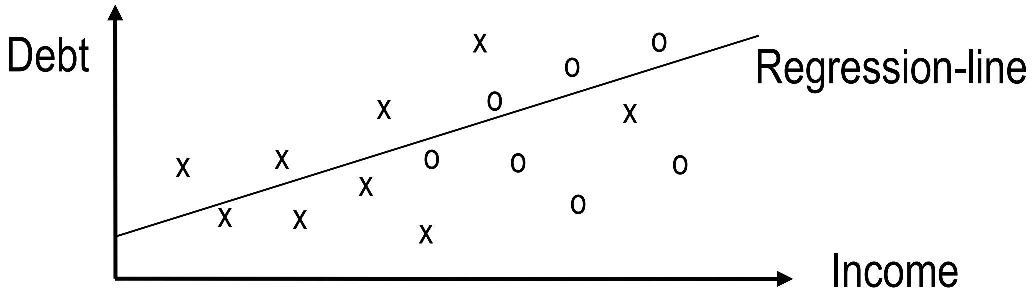
Regression is learning a function that maps a data item into a real-valued prediction variable.

Applications

Automated prediction of numerical properties (e.g. customer age) of objects in large market databases

Data Mining Tasks

Regression



Examples

• Predicting consumer demand for a new product as a: function of advertising expenditure.

• Fitting the total debt as a linear function of income.

Data Mining Tasks

Data mining primary tasks

Affinity grouping involves methods for finding associations between groups of variables.

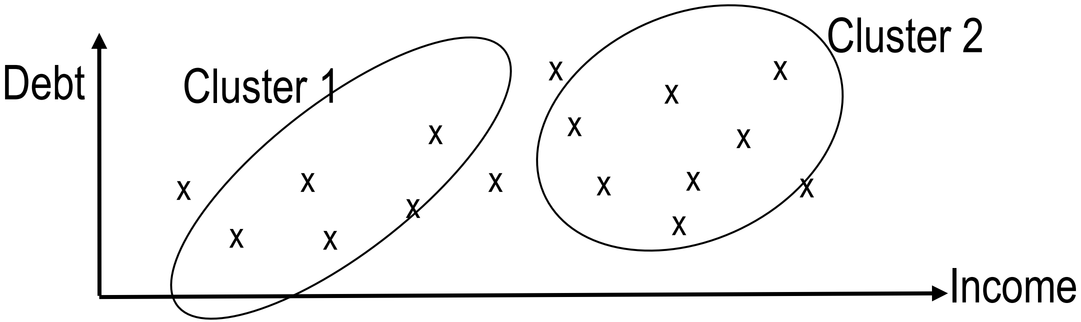
Examples:

- Association rules, i.e. expressions of the type $X \rightarrow Y$, where X and Y are sets of binary features. The intuitive meaning of such a rule is that transactions of the database which contain X tend to contain Y.

“98% of customers that purchase tires and automobile accessories also have automotive services carried out”

Data Mining Tasks

Clustering



Examples:

• Discovering homogeneous sub-populations of consumers in marketing databases.

• Identification of sub-categories of spectra from infra-red sky measurements.

• Loan data analysis.

Data Mining Tasks

Clustering

Closely related to clustering is the task of probability density estimation, which consists of techniques for estimating from data the joint multi-variate probability density function of all of the variables/fields in the database.

Data Mining Tasks

Data mining primary tasks

Summarization (or description or profiling) involves methods for finding a compact description for a subset of data.

Examples:

- Generating a number of statistics (e.g., mean and standard deviations) of all variables. Summarization is typical of periodic reporting activities.

Data Mining Tasks

Change and deviation detection

Change and deviation detection focuses on discovering the most significant changes in the data from previously measured or normative values.

It consists of finding a model that describes significant dependencies between variables.

Data Mining Tasks

Change and deviation detection

Examples:

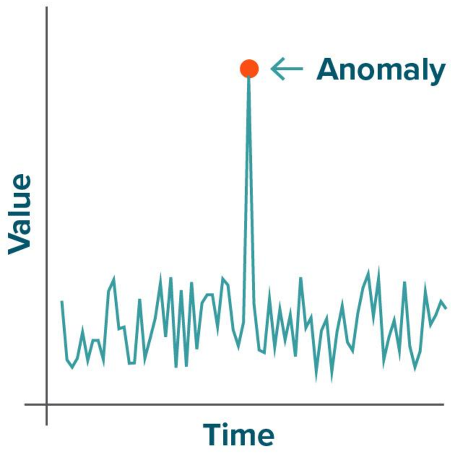
- Outlier detection, i.e. identification of "strange examples" that may be due to measurement errors.
- Emerging patterns, i.e. patterns that appear to be frequent in a class and unfrequent in other class(es). Provide a characterization of a class.

Data Mining Tasks

Change and deviation detection

Applications:

- Fraud detection in the use of credit cards, insurance claims, and telephone cards.
- Quality control.
- Defects tracing.
- Loan data analysis



Business Understanding

Produce Project Plan

At the end of the business understanding phase, we describe the intended plan for achieving the data mining goals, and thereby achieving the business objectives.

OUTPUT:

- the project plan, which specifies the set of steps for the rest of the project, along with duration, required resources, inputs and outputs, and dependencies.

Customer Vulnerability Analysis (CVA)

In the customer vulnerability analysis the problem is the development of customer *vulnerability models* that help predict consumer loyalty levels to a particular product or class of products.

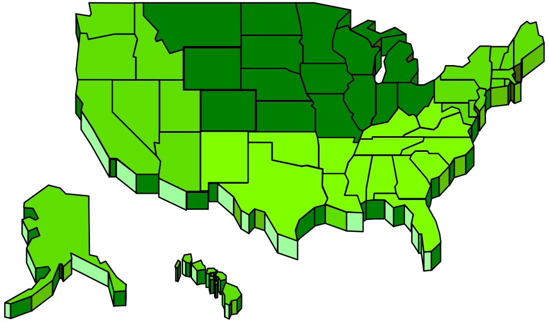
Such models are important to packaged goods companies, for example, who aim to secure or increase market share for their new or existing products. Predictions from the vulnerability models help these companies in planning marketing strategies

- In-store promotions
- Direct mail campaigns
- Advertising of discount coupons

for the launch or promotion of their products.

Customer Vulnerability Analysis (CVA)

Business objective: develop a marketing strategy to maintain the current market share for a particular brand and size (12-oz. can) of frozen orange juice in the Midwestern U.S. during the months of November, December and January.

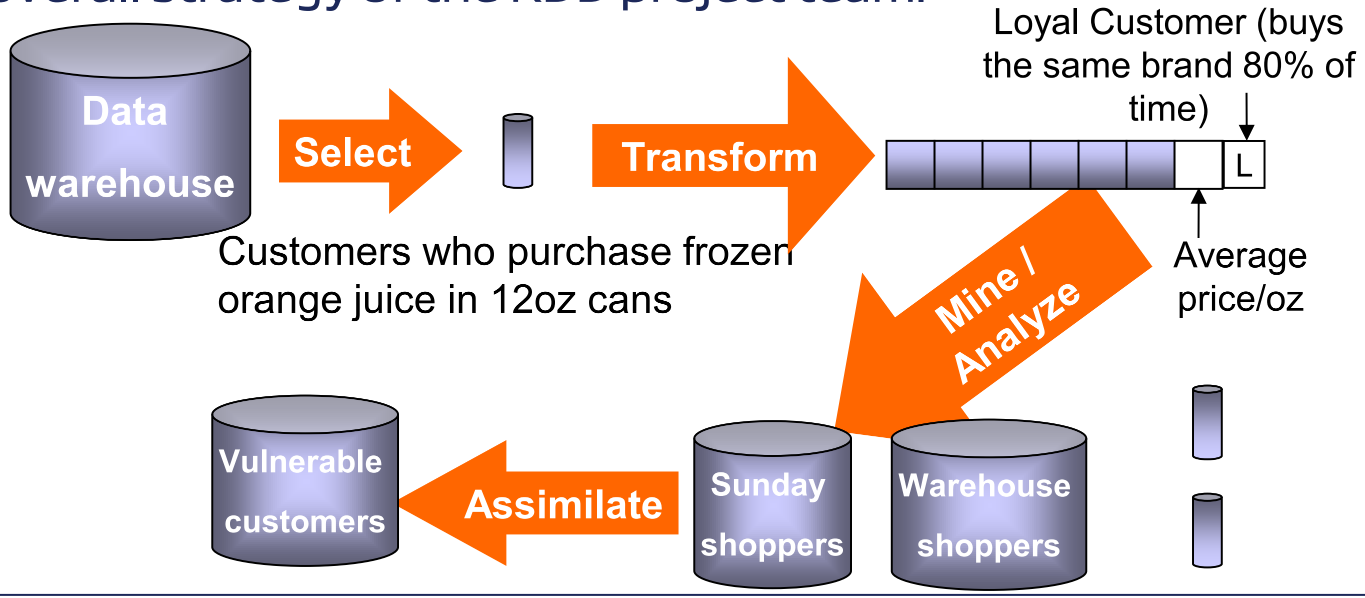


The company would use a combination of marketing strategies, one of which was to be a direct mail campaign aimed at customers who were perceived to be vulnerable.

Customer Vulnerability Analysis (CVA)

Key question: *Which households should be included in the direct mail campaign for the effort to be successful?*

The overall strategy of the KDD project team.



Customer Vulnerability Analysis (CVA)

The team used syndicated data as a basis to build models of the typical buying patterns of both vulnerable and loyal customers.

Once these behavioral rules were visible and understood, the team segmented the customer database into homogeneous groups for targeting by different marketing campaigns.

syndicated data= dati consorziali.